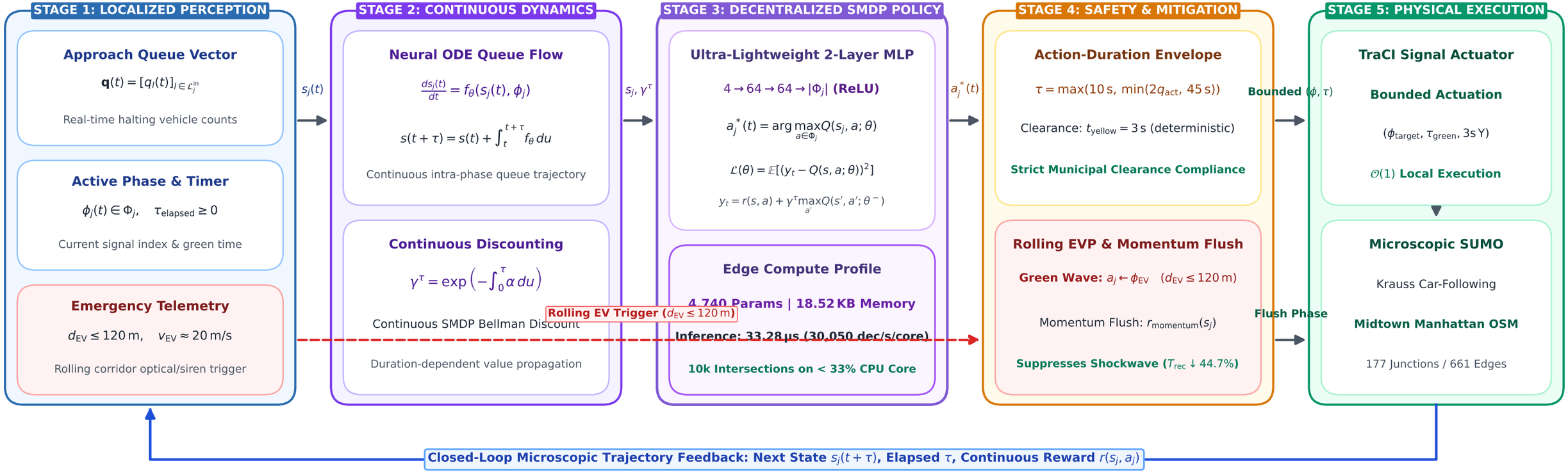

STAGE 1: LOCALIZED PERCEPTION
Approach Queue Vector
$\mathbf{q}(t) = [q_l(t)]_{l \in \mathcal{L}_j^{\text{in}}}$
Real-time halting vehicle counts
Active Phase & Timer
$\phi_j(t) \in \Phi_j, \quad \tau_{\text{elapsed}} \geq 0$
Current signal index & green time
Emergency Telemetry
$d_{\text{EV}} \leq 120\,\text{m}, \quad v_{\text{EV}} \approx 20\,\text{m/s}$
Rolling corridor optical/siren trigger
$s_j(t)$
STAGE 2: CONTINUOUS DYNAMICS
Neural ODE Queue Flow
$\frac{ds_j(t)}{dt} = f_\theta(s_j(t), \phi_j)$
$s(t+\tau) = s(t) + \int_t^{t+\tau} f_\theta\, du$
Continuous intra-phase queue trajectory
Continuous Discounting
$\gamma^\tau = \exp\left(-\int_0^\tau \alpha\, du\right)$
Continuous SMDP Bellman Discount
Duration-dependent value propagation
s_j, γ^τ
STAGE 3: DECENTRALIZED SMDP POLICY
Ultra-Lightweight 2-Layer MLP
$4 \to 64 \to 64 \to |\Phi_j|$ (ReLU)
$a_j^*(t) = \arg\max_{a \in \Phi_j} Q(s_j, a; \theta)$
$\mathcal{L}(\theta) = \mathbb{E}[(y_t - Q(s,a;\theta))^2]$
$y_t = r(s,a) + \gamma^\tau \max_{a'} Q(s', a'; \theta^-)$
Edge Compute Profile
4,740 Params | 18.52 KB Memory
Inference: 33.28 µs (30,050 dec/s/core)
10k Intersections on < 33% CPU Core
Rolling EV Trigger ($d_{\text{EV}} \leq 120$ m)
$a_j^*(t)$
STAGE 4: SAFETY & MITIGATION
Action-Duration Envelope
$\tau = \max(10\,\text{s}, \min(2q_{\text{act}}, 45\,\text{s}))$
Clearance: $t_{\text{yellow}} = 3$ s (deterministic)
Strict Municipal Clearance Compliance
Rolling EVP & Momentum Flush
Green Wave: $a_j \leftarrow \phi_{\text{EV}}$ ($d_{\text{EV}} \leq 120$ m)
Momentum Flush: $r_{\text{momentum}}(s_j)$
Suppresses Shockwave ($T_{\text{rec}} \downarrow 44.7\%$)
Bounded (ϕ, τ)
Flush Phase
STAGE 5: PHYSICAL EXECUTION
TraCI Signal Actuator
Bounded Actuation
(ϕ_{target}, τ_{green}, 3s Y)
$\mathcal{O}(1)$ Local Execution
Microscopic SUMO
Krauss Car-Following
Midtown Manhattan OSM
177 Junctions / 661 Edges
Closed-Loop Microscopic Trajectory Feedback: Next State $s_j(t+\tau)$, Elapsed τ, Continuous Reward $r(s_j, a_j)$